

2. Overall Description

a. User Classes and Characteristics

Microgrid Operators

Microgrid operators will use the dashboard to monitor grid conditions, energy flow, node status, and possible overload situations.

Power Utility Engineers

Power utility engineers will analyze grid performance, predicted demand spikes, capacity constraints, and anomaly conditions.

Smart City Infrastructure Managers

Smart city managers will use the system to monitor simulated energy infrastructure and evaluate grid behavior under different shock scenarios.

System Administrators

Administrators may manage system configuration, simulated grid parameters, user access, and system monitoring.

b. User Needs

The users require a system that can:

- Monitor grid conditions through an interactive dashboard.
- Understand the current state of grid nodes and substations.
- Visualize energy movement across the grid.
- Identify potential overload situations.
- Predict future demand spikes.
- Receive warnings about abnormal grid conditions.
- Test the effect of simulated grid events.
- Ensure AI recommendations follow physical grid constraints.
- Analyze grid behavior without requiring physical electrical hardware.

c. Operating Environment

GridPulse will operate as a web-based full-stack application.

Frontend Environment

- React / Next.js
- D3.js
- Recharts
- Tailwind CSS
- Modern web browser

Backend Environment

- Python
- FastAPI
- Celery
- Redis

Database Environment

- PostgreSQL
- TimescaleDB extension

Data Processing and AI Environment

- Pandas
- NumPy
- Scikit-Learn
- XGBoost / LSTM

Simulation Environment

A Python-based grid simulator will generate mock IoT telemetry for the simulated 20-node grid. WebSockets may be used for transmitting telemetry to the application.

d. Constraints

The major constraints of GridPulse are:

- The system will operate on a simulated grid rather than a physical power grid.
- The initial system scope will be limited to approximately 20 nodes.
- Real-world electrical infrastructure will not be directly controlled.
- AI recommendations must pass predefined safety rules.
- Cable and node capacity limits must not be exceeded.
- The system depends on simulated telemetry for testing and demonstration.
- The forecasting accuracy will depend on the quality and quantity of available training data.

e. Assumptions

The project assumes that:

- Simulated telemetry will represent basic real-world grid behavior.
- Grid nodes and cable capacities will be predefined.
- Sufficient historical telemetry will be available for AI model training.
- Users have access to a modern web browser.
- The backend and database services will be available during system operation.
- AI predictions are advisory and will not directly control physical grid equipment.
- The simulated grid will contain a maximum of approximately 20 nodes within the initial project scope.